

The idea that AI art produces more pollution needs to stop.

Posted by u/ScarletIT • 0 points • 121 comments

There is this argument that gets parroted around about the carbon footprint of image generation.

There is a significant amount of energy invested on developing an AI system, that does produce emissions. Big computers, multiple computers working constantly to process training. Sure.

But that happens once, and then stops.

I see many poorly understood arguments from antis convinced that every single image produced consumes an ungodly amount of energy and therefore causes a significant impact on carbon footprint.

A PC cannot really be fed much more energy than average.

If I use stable diffusion locally, what my pc consumes is the impact that the image generation is going to have, and is really not different than me listening to music on spotify.

You may argue "but people use services online to generate images and you don't know how much computing power and how many computers are behind that" sure... but you know, they do exactly what they do with my one pc.

Unlike the argument when it was applied to crypto when you actually had millions of individuals mining crypto day and night, sometimes on multiple pcs at the same time, the environmental impact of AI art is turning on the pc, running the model for a few minutes to generate an image, and then turning off.

Technically speaking, photoshop has a way bigger impact as you need to spend longer time on it to produce the same picture.

Your pc being on using photoshop, being on generating AI art and being on checking e-mails are not significantly different as an impact.

The impact, the energy consuming process, is the building of the algorithms.

But that is a one and done.

It's not something that every average joe who uses AI art does.

The guys at stability AI are doing it, that's it.

People who create a model consume energy as the process is long.

But that long process is usually a few hours long. If you are playing a videogame for those few hours, your impact is basically the same.

We need to kill this rumor that AI Art magically turns a PC into this Captain planet villain contraption that spews toxic gas in the air.

If you work in an office for an environmental foundation and there are 10 pcs there running for 8 hours a day, then you go home and make AI art for an hour, your office generated 80 times more pollution than you.

There is nothing magically toxic about creating AI art. Is just computers running at their usual pace.

There is an upfront energy cost in training, but lots of articles fallaciously multiply that cost per user and come up with apocalyptic numbers.

That's not how it works.

Comments

u/No-Range1803 • 1 points • 2024-08-07 17:18 UTC

Energy consumption isn't about "how hot a computer runs" or any perception that is similar to that.

The broader concern is impact to the grid, carbon that is related to model training, and server capacity. Someone else notes as an example they don't generate one AI image, they generate 30. The sheer mass of generation across many users, feeding back into the grid, is where a potential problem exists.

The true problem isn't that we need more energy to do things we want. It is the source of the energy. As long as grids are run off fossil fuels, emissions will be a problem. We don't have sufficient ways of transitioning our grids off fossil fuels, it is imperative we do it sooner as the AI demands are growing. That's not the only demand - electrifying buildings, commercial production, agriculture and transit of all kinds is necessary. This strains grids further.

The caveat to me is that we may need to prioritize our energy use (this happens already with rolling blackouts for example).

TL;dr - AI energy use is a problem, but the real issue is transitioning grids off fossil fuels which needs to happen soon so we can meet electricity needs (and no, nuclear is not the answer either).

u/Sbarty • 1 points • 2024-08-06 22:35 UTC

Carbon footprint is a pretty dumb metric anyways. If the entire west got their carbon footprint to 0 tomorrow, India, China, and a plethora of overpopulated third world countries still exist.

What's the point? It's a drop in the ocean and doesn't get us even close to reversing what's been done, let alone slowing down the onset of worse problems.

u/ScarletIT • 1 points • 2024-08-06 22:40 UTC

Just saying.

But AI arguments do not affect the west only.

u/Wave_Walnut • 1 points • 2024-08-04 05:06 UTC

An inconvenient truth:

Google and Microsoft consumed more electricity than 100-plus countries, the 'bad and good news' in these numbers

<https://www.msn.com/en-in/news/world/google-and-microsoft-consumed-more-electricity-than-100-plus-countries-the-bad-and-good-news-in-these-numbers/ar-BB1qbRih>

u/ScarletIT • 1 points • 2024-08-04 06:56 UTC

Not more. Than 100 countries combined.

If it was a country, one if those companies would consume more than 100 of them, and less than 95.

So what countries does it place itself around?

Azerbaijan and Libya.

Yeah... I am sorry, I do expect a company like microsoft to consume roughly what those countries consume.

u/Doctor_Amazo • 1 points • 2024-08-03 16:22 UTC

Why? It's a fact. The carbon footprint to run AI is massive.

u/realechelon • 2 points • 2024-08-03 14:06 UTC

I went into this argument at length before:

[https://www.reddit.com/r/aiwars/comments/1dmkpby/the_environmental_argument_against_ai_art_is_bogus/]
(https://www.reddit.com/r/aiwars/comments/1dmkpby/the_environmental_argument_against_ai_art_is_bogus/)

The antis who replied were mostly completely uninterested in facts.